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Intensified Indonesian Throughflow driven by La Niña-like state and stronger global thermohaline circulation during the last interglacial

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ABSTRACT

The Indonesian Throughflow (ITF) is a key part of the global thermohaline circulation. However, the mechanisms and environmental impacts of the ITF in the geological past remain inadequately investigated. Here, we present model outputs from a set of Last Interglacial (LIG) snapshot simulations carried out by CAS-FGOALS (the Chinese Academy of Sciences Flexible Global Ocean–Atmosphere–Land System model) under the protocol of PMIP (Paleoclimate Modeling Intercomparison Project). Compared to the piControl simulations (the annual mean ITF flux was 18.46 Sv), an annual mean ITF flux increase of about 30.6%–35.9% was found in the early LIG snapshots (24.11–25.08 Sv). This enhancement of the ITF was closely correlated with a La Niña-like state and increased Pacific upwelling. During the LIG, an increased zonal sea surface temperature gradient and intensified easterlies in the tropical Pacific acted as positive feedback loop, resulting in a La Niña-like state with rising sea level in the western Pacific. As a result, the sea level gradient between the tropical western Pacific and the tropical eastern Indian Ocean increased, enhancing the pressure contrast between the two basins and ultimately strengthening the ITF volume transport. Additionally, increased North Atlantic Deep Water formation during the LIG led to enhanced Pacific deep upwelling, which contributed to the strengthening of ITF transport. Comparisons between models and proxies further support a La Niña-like state, enhanced Atlantic Meridional Overturning Circulation, and intensified ITF transport during the early LIG.

1. Background

The global thermohaline circulation is a fundamental component of the Earth's climate system, redistributing heat, freshwater, and nutrients across ocean basins (Fig. 1A; Broecker, 1991; Talley, 2013). This global overturning is primarily characterized by two main overturning cells associated with the formation of North Atlantic Deep Water (NADW) and Antarctic Bottom Water (AABW). NADW forms in the North Atlantic, flows southward at depth, and joins the Antarctic Circumpolar Current (ACC). Subsequently, while a minor fraction of NADW join the more northerly upwelling of Indian Deep Water (IDW) and Pacific Deep Water (PDW) (Lumpkin and Speer, 2007), the vast majority upwells south of the ACC, where intense surface cooling and brine rejection convert it into AABW (Talley, 2013). AABW flows northward into all three major basins. In the Atlantic Ocean, it upwells and merges with the southward-flowing NADW. In the Pacific and Indian Oceans, most AABW transforms into IDW and PDW and returns to the Southern Ocean, where it either contributes to new AABW formation or feeds the northward surface water flux that eventually supplies NADW. The remaining fraction of IDW and PDW continues upwelling into the thermocline. This upwelled PDW then transits the Indonesian Throughflow (ITF) into the Indian Ocean, merging with the upwelled IDW to supply the Agulhas Current, the majority of which ultimately flows into the Atlantic (Talley, 2013).

As a key component of the global thermohaline circulation, the ITF connects the Pacific and the Indian Ocean, regulates the heat and freshwater budgets between them (Gordon, 2005), and ultimately brings waters from the Pacific into the North Atlantic (Beal et al., 2011; Gordon, 2005). Today, the ITF transports ~15 Sv of water from the Pacific into the Indian Ocean (Sprintall et al., 2009). This substantial volume transport is accompanied by significant heat transport (Godfrey, 1996; Vranes et al., 2002), potentially impacting the El Niño-Southern Oscillation (ENSO) and monsoon phenomena (Godfrey, 1996; Gordon, 2005; Webster et al., 1998). Recent studies also suggest that the ITF may have contributed to the global warming hiatus during the first decade of the 21st century by transferring heat from the tropical Pacific

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to the Indian Ocean (England et al., 2014; Lee et al., 2015; Li et al., 2017).

However, despite the important role of the ITF in the climate system, its behavior and environmental effects in the context of future global warming are not fully understood. While climate models consistently indicate a 21st century ITF weakening (20-32% reduction under RCP8.5 or 47% reduction under abrupt quadrupling of CO_2) (Feng et al., 2017; Peng et al., 2023; Sen Gupta et al., 2016; Stellema et al., 2019), observations since the 1990s reveal an intensification of the ITF (about 1.33–3.7 Sv/decade increase) (Guo et al., 2023; Hu and Sprintall, 2017) coinciding with an intensified Pacific Walker circulation (L'Heureux et al., 2013). This discrepancy between model projections and observational evidence highlights key uncertainties in understanding ITF dynamics.

Today's ITF is fed by both North Pacific and South Pacific waters, with an additional contribution from the South China Sea (Fig. 1B). The surface to upper thermocline waters mainly originate from the Mindanao Current (North Pacific), and enter via the Makassar Strait (Gordon,

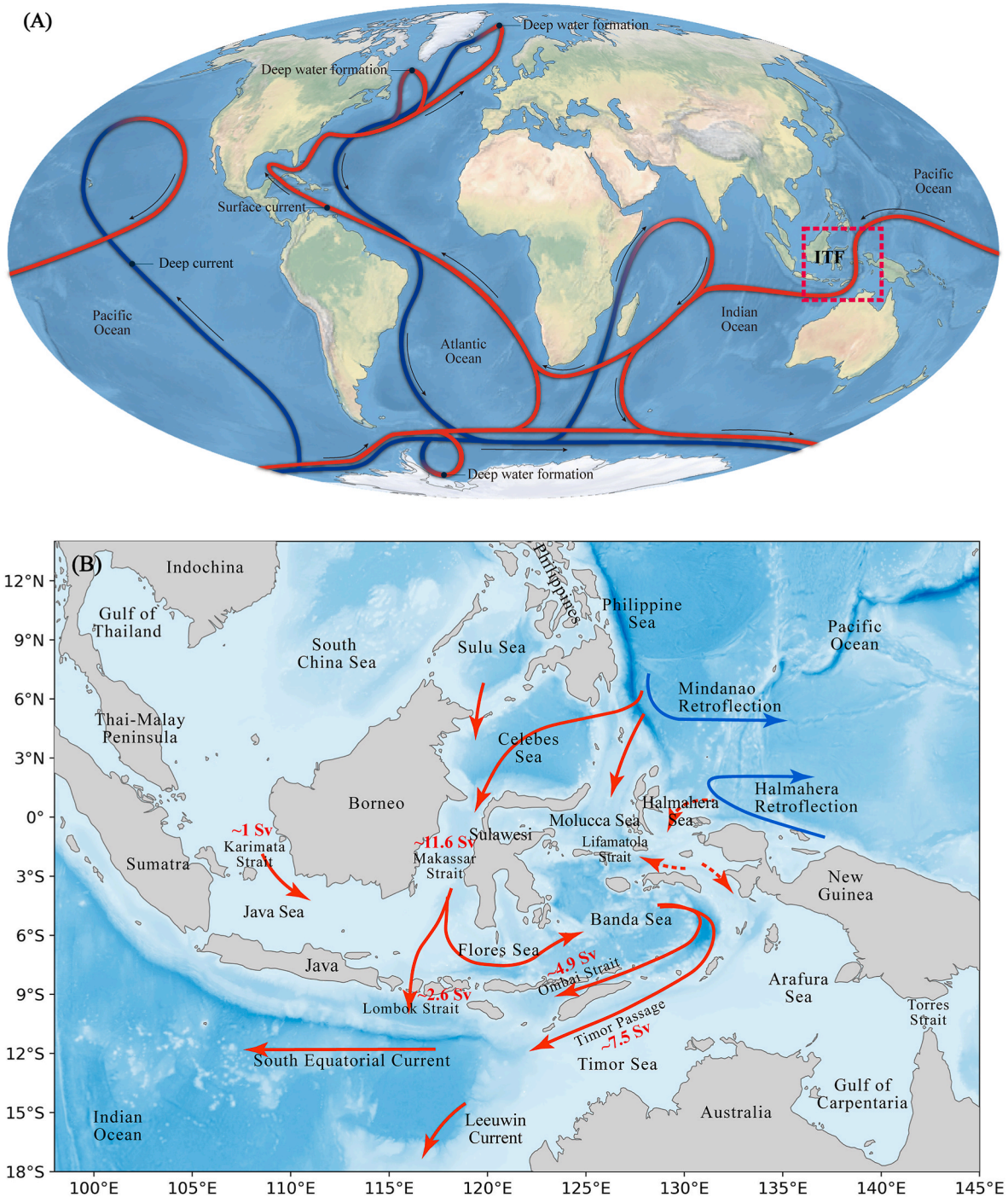


Fig. 1. (A) Conceptual diagram illustrating the global ocean overturning circulation. Red indicates warm surface currents, while blue indicates cold deep currents. The ITF area is marked by the red dashed-line box. Adapted from Pravettoni (2006). (B) The Indonesian Seas and pathways of the ITF. The ITF derives from the Mindanao and Halmahera retroflections. Most of the ITF inflow passes through the Makassar Strait, while the inflow entering via the Halmahera Sea and Lifamatola Strait is not well constrained. A small portion of the ITF enters the Indian Ocean via the shallow Lombok Strait, while the main ITF outflow travels through the Ombai and Timor Straits. Red lines denote the ITF pathways. The schematic diagram is adapted from Sprintall et al. (2014).

2005; Sprintall et al., 2009). The total transport through the Makassar Strait was estimated to be 11.6 ± 3.3 Sv, as observed by the International Nusantara Stratification and Transport (INSTANT) program from late 2003 to early 2007 (Gordon et al., 2008; Sprintall et al., 2009). While most of these waters then enter the Flores Sea and flow eastward to the Banda Sea, a portion directly enters the Indian Ocean via the Lombok Strait (Gordon, 2005; Sprintall et al., 2009). Additionally, a small part of the North Pacific surface water enters the Banda Sea through the Lifamatola Strait (Gordon, 2005; Sprintall et al., 2009). The lower thermocline and deeper components mainly originate from the South Pacific and enter via the Halmahera Sea and Lifamatola Strait (Gordon, 2005; Sprintall et al., 2009); however, the transport of these components is not well constrained and carries large uncertainties (Gordon et al., 2010; Sprintall et al., 2014). Additional smaller surface waters enter from the South China Sea via the Karimata Strait in the Java Sea (~1 Sv) (Tozuka et al., 2007; Sprintall et al., 2009). The Torres Strait between Australia and New Guinea is too shallow (~10 m) to allow significant throughflow (Gordon, 2005).

After traveling from the Pacific Ocean to the Indonesian Seas, the waters undergo intense mixing in the Banda Sea due to tidal forces and wind-driven upwelling (Gordon, 2005; Sprintall et al., 2009; Feng et al., 2018). These waters then exit into the Indian Ocean through three major straits: the Lombok Strait, Ombai Strait, and Timor Passage (Gordon, 2005; Sprintall et al., 2009), with total volume transports of 2.6 Sv, 4.9 Sv, and 7.5 Sv observed by INSTANT, respectively (Sprintall et al., 2009). After exiting into the Indian Ocean, the ITF waters join the South Equatorial Current and undergo complex water mass transformation processes, with the majority eventually feeding into the Agulhas Current (Godfrey, 1996; Feng et al., 2018). A smaller portion of the ITF waters flows southward along the western Australian coast, constituting a source for the Leeuwin Current (Feng et al., 2018).

So far, ITF variations have been primarily linked to the Asian-Australian monsoon system (Sprintall et al., 2009; Ding et al., 2023), the ENSO state (Wyrtki, 1987; Sprintall et al., 2009; Zhang et al., 2018), the Indian Ocean Dipole (Feng et al., 2018), and sea level changes (Pang et al., 2021; Ding et al., 2023). Yet, little is known about the influence of deep ocean circulation on the ITF. While future climate projections suggest a strong link of ITF variations to deep Pacific upwelling (Feng et al., 2017; Sen Gupta et al., 2016; Stellema et al., 2019) and the remote Atlantic Meridional Overturning Circulation (AMOC) effect (Peng et al., 2023; Sun and Thompson, 2020), this hypothesis lacks constraints from proxy records and geological analogues.

The Last Interglacial (LIG, 130–115 ka), also known as Marine Isotope Stage 5e (MIS 5e) or the Eemian, was the penultimate interglaciation. With abundant proxy records and well-constrained boundary conditions, the LIG stands out as an excellent period to study climate interactions in a warmer-than-present world. As one of the warmest interglacials over the past 800 ka (Clark et al., 2024; Jouzel et al., 2007), the LIG had a peak global mean sea level (GMSL) traditionally estimated at 6–9 m above present (Dutton et al., 2015; Kopp et al., 2009), though recent assessments suggest a much lower peak of 1.2-5.3 m (Dyer et al., 2021). It also featured a peak global mean annual SST (sea surface temperature) of 0.5 ± 0.3°C warmer than the 1870-1889 mean (Hoffman et al., 2017). This warmth is primarily attributed to changes in the seasonal and latitudinal distribution of incoming solar radiation at the top of the atmosphere (Berger and Loutre, 1991).

Proxy-based reconstructions generally indicate ITF intensification during the LIG (Holbourn et al., 2011; Pang et al., 2021) due to changes in the monsoon system, sea level, and global thermohaline circulation (Xu et al., 2006; Holbourn et al., 2011; Pang et al., 2021; Ding et al., 2023). Orbitally paced monsoon wind forcing, together with sea-level–controlled gateway geometry, modulated the ITF's magnitude and vertical structure (Xu et al., 2006; Pang et al., 2021; Ding et al., 2023), favoring a more thermocline-dominated throughflow export during the LIG (Xu et al., 2006; Holbourn et al., 2011; Pang et al., 2021). In addition, Holbourn et al. (2011) inferred that variations in the deeper ITF below 500 m may be related to changes in global thermohaline circulation intensity. However, existing studies remain qualitative (e.g., "stronger"/"weaker"), lacking physically grounded, quantitatively testable mechanistic interpretations.

Here, we present results from LIG climate simulations using a state-of-the-art coupled climate model, providing new insights into ITF dynamics and comparisons with published proxy records. These results reveal the underlying mechanisms of the ITF behavior in a warmer-than-present world. Furthermore, we compare the ITF during the LIG with that in future climate projections.

2. Methods

2.1. Model introduction

We used the state-of-the-art coupled climate model FGOALS-g2 (Flexible Global Ocean–Atmosphere–Land System Model Grid-point Version 2) to carry out a series of LIG snapshot simulations. FGOALS-g2 has been used in the CMIP5 (fifth Coupled Model Inter-comparison Project) and IPCC (Intergovernmental Panel on Climate Change) AR5 (the Fifth Assessment Report). FGOALS-g2 showed improvements in cloud-radiation processes, the Asian monsoon, the ENSO phenomenon, and sea ice. Enhanced cloud fraction parameterization improves the accuracy of simulations for both shortwave and longwave cloud radiative forcing, with results approaching the CMIP5 multimodel mean. Together with improved thermocline feedbacks (Yu et al., 2013; Chen et al., 2016), these advancements enable FGOALS-g2 to become one of the best CMIP5 models for reproducing ENSO behaviors (Bellenger et al., 2014; Zhou et al., 2018). Furthermore, FGOALS-g2 excels in simulating the western North Pacific anomalous anticyclone, which is critical for East Asian summer monsoon variability (Zhou et al., 2018). The model also shows systematic improvements in simulating sea ice seasonality and extent compared to its predecessor, though biases exist in concentration distributions and regional heat transport (Lin et al., 2013). Building on these strengths, FGOALS-g2 has demonstrated strong performance in paleoclimate reconstruction, modern climate diagnostics, and future climate projections (Chen et al., 2016; Wang et al., 2013; Zheng and Yu, 2014; Shi et al., 2024; Zhang et al., 2025).

FGOALS-g2 contains four component models: the atmospheric circulation model (GAMIL2), the ocean circulation model (LICOM2), the land surface process model (CLM3), and the sea ice model (CICE4-LASG). These four component models are coupled through a flux coupler CPL6 (Craig et al., 2005).

The Grid-point Atmospheric Model of LASG/IAP Version 2 (GAMIL2) has a horizontal resolution of 2.8°, employing a weighted equal-area grid poleward of 65.58° N/S. It features 26 vertical layers, extending up to 2.194 hPa. The dynamical core utilizes a finite difference scheme to solve the hydrostatic primitive equations, ensuring the conservation of mass and effective energy (Wang et al., 2004). It also employs a two-step shape-preserving advection scheme for the moisture equation (Yu, 1994). A detailed description is provided in Li et al. (2013). The Community Land Model Version 3 (CLM3) (Oleson et al., 2010) operates on the same grid as GAMIL2. CLM3 is built upon several foundational models: the Biosphere–Atmosphere Transfer Scheme (BATS), the IAP Land Surface Model (IAP94), and the Land Surface Model (LSM). The LASG/IAP Climate System Ocean Model Version 2 (LICOM2) features a horizontal resolution of 1° × 1°, with a 0.5° meridional resolution in the tropics. It includes 30 vertical layers, with a 10 m resolution in the upper 150 m. Improvements have been made to various physical processes, such as the mixing schemes (Canuto et al., 2001, 2002; Liu et al., 2012) and the solar penetration scheme (Lin et al., 2007, 2011). Improved by LASG, CICE4-LASG is an enhanced version of CICE4.0 (Los Alamos sea ice model version 4.0). CICE4-LASG employs elastic-viscous-plastic ice dynamics (Hunke and Dukowicz, 1997), an incremental remapping-based advection scheme (Dukowicz and Baumgardner, 2000), and an energy-conserving thermodynamics model (Bitz and

Lipscomb, 1999).

2.2. Experiment design

The LIG represents Earth's most recent natural warm analog, characterized by distinct latitudinal and seasonal distributions of incoming solar radiation driven by Earth's orbital parameters. The PMIP3 protocol established four experimental configurations for the LIG, including three Eemian simulations (130 ka, 128 ka, and 125 ka) with significant orbital parameter variations, and one experiment targeting the onset of glacial inception (115 ka) (Lunt et al., 2013).

The PMIP3 protocol prioritizes orbital forcing as the LIG's dominant climate driver as calculated by Berger (1978). Other boundary conditions - such as vegetation, aerosols, solar constant, ice sheets, topography, and coastlines - were maintained at the pre-industrial (PI) levels to isolate orbital effects. Key changes of the orbital forcing included increased obliquity, enhanced eccentricity, and coinciding of boreal summer with perihelion (Berger, 1978). Greenhouse gas concentrations during the LIG were near PI levels based on ice core records (Loulergue et al., 2008; Lüthi et al., 2008; Spahni et al., 2005), with CO_2 set to 257 ~276 ppm in the LIG experiments (Lüthi et al., 2008). Although the LIG featured expanded boreal forests (Castañeda et al., 2009; Crump et al., 2021) and elevated sea level (Dutton et al., 2015; Kopp et al., 2009), models retained PI vegetation and coastlines due to limited paleovegetation reconstructions and to prevent other effects from obscuring the orbital signals.

Following the PMIP3 protocol, five snapshot simulations were carried out using FGOALS-g2, corresponding to 130ka, 128ka, 125ka, 115ka, and PI (Table 1). All simulations ran sufficiently long to ensure the system reached equilibrium. The PI simulation was integrated for 900 model years, the 115ka experiment for 1600 model years, and the 125-130ka experiments for 1900 model years each. We used the averages from the last 50 years of each simulation for further analysis.

2.3. Calculation of volume transport

Considering the main pathways of the ITF, we divided the Indonesian Seas into eight sections (Fig. 2). These sections include: ①the transect along 114°E from Australia to Indonesia, representing the total ITF volume transport; ②the Ombai Strait; ③the Timor Passage; ④the Torres Strait; ⑤the Makassar Strait; ⑥the Lifamatola Strait; ⑦the Halmahera Sea; and ⑧the Karimata Strait. For each section, we calculated the volume transport for the full depth, the upper 354 m, and the upper 1021 m to identify at which depth the transport changed the most.

Volume transport is calculated by integrating the velocity normal to the section over the cross-sectional area, from the surface to the required depth, as shown in Equation (1). Where F denotes the volume transport (m^3/s), v is the velocity normal to the section (m/s), and dA is the differential element of the sectional area A(m^2). In practice, the section can be divided into discrete segments, and the transport is approximated by summing the product of velocity and area for each segment using Equation (2). Where n is the total number of segments, v_i is the velocity of the i-th segment, and A_i is the area of the i-th segment.

$$F = \int_A v \mathrm{d}A \tag{1}$$

$$F \approx \sum_{i=1}^{n} v_i \cdot A_i \tag{2}$$

2.4. Proxy calibration

Globigerinoides ruber calcifies in near surface conditions (0–50 m), recording mixed-layer temperatures, while *Pulleniatina obliquiloculata* calcifies in deeper waters (50–100 m), reflecting thermocline temperatures (Anand et al., 2003). The temperature gradient between these two species ($\Delta T_{(G.\ ruber\text{-}P.\ obliquiloculata)}$, or ΔT for simplicity) could be used to suggest the depth of the thermocline (DOT). This approach has been a well-established and widely accepted proxy for thermocline depth in paleoceanography for over two decades (Andreasen and Ravelo, 1997; Pflaumann and Jian, 1999; Jian et al., 2000, 2022; Xu et al., 2006, 2008; Holbourn et al., 2011; Ding et al., 2023). Specifically, an increased ΔT indicates a shoaling thermocline, whereas a decreased ΔT suggests a deepening thermocline.

The tropical oceans closely resemble a two-layer system, characterized by a strong correlation between sea surface height (SSH) and DOT, suggesting that the dynamics of the upper ocean has only one vertical baroclinic mode (Rebert et al., 1985). The pressure gradient induced by the SSH gradient reaches its maximum at the sea surface and gradually decreases with depth, approaching zero near the permanent thermocline without reversing sign. While wind stress directly forces Ekman currents within the surface boundary layer (in the upper 50–100m), the currents beneath it are primarily governed by the pressure gradient because the turbulent stress can be ignored outside of the boundary layer. Consequently, depth-integrated transports in the upper ocean, such as the ITF, are primarily driven by the pressure gradient and, by extension, the SSH gradient. The relationship between ITF transport and the inter-basin pressure gradient was first established by Wyrtki (1987). Subsequent studies have quantitatively explored this relationship using model outputs (Potemra, 2005) or observational data (Sprintall and Révelard, 2014). For instance, Sprintall and Révelard (2014) derived ITF transport estimates from regional sea level anomalies using linear regression, obtaining significant correlations of ~0.8 with INSTANT observations at the interannual time scale. Therefore, the reconstructed DOT can be used to infer the total transport of the ITF.

We collected Mg/Ca records of surface- and subsurface-dwelling foraminifera from seven sites in the Indonesian Seas (Fig. 2; Bolliet et al., 2011; Ding et al., 2023; Hollstein et al., 2020; Jian et al., 2020,

Table 1
Boundary conditions for the set of LIG snapshot simulations.

		PI	130ka	128ka	125ka	115ka
Orbital parameters	ecc	0.016724	0.038209	0.039017	0.040013	0.041421
	Obl	23.446°	24.242°	24.131°	23.798°	22.405°
	Longitude of perihelion	102.04°	228.32°	259.65°	307.14°	110.88°
Date of vernal equinox		March 21 at noon	March 21 at noon	March 21 at noon	March 21 at noon	March 21 at noon
Trace gases	CO_2	280 ppm	257 ppm	275 ppm	276 ppm	273 ppm
	CH_4	760 ppb	512 ppb	709 ppb	640 ppb	472 ppb
	N_2O	270 ppb	239 ppb	266 ppb	263 ppb	251 ppb
	CFC	0	0	0	0	0
	O_3	Modern −10 DU	Same as PI	Same as PI	Same as PI	Same as PI
Aerosols		Same as PI	Same as PI	Same as PI	Same as PI	Same as PI
Solar constant		1365 W/m^2	1365 W/m^2	1365 W/m^2	1365 W/m^2	1365 W/m^2
Vegetation		Interactive	Pre-industrial fixed	Pre-industrial fixed	Pre-industrial fixed	Pre-industrial fixed
Ice sheets		Present day	Same as PI	Same as PI	Same as PI	Same as PI
Topography and coastlines		Present day	Same as PI	Same as PI	Same as PI	Same as PI

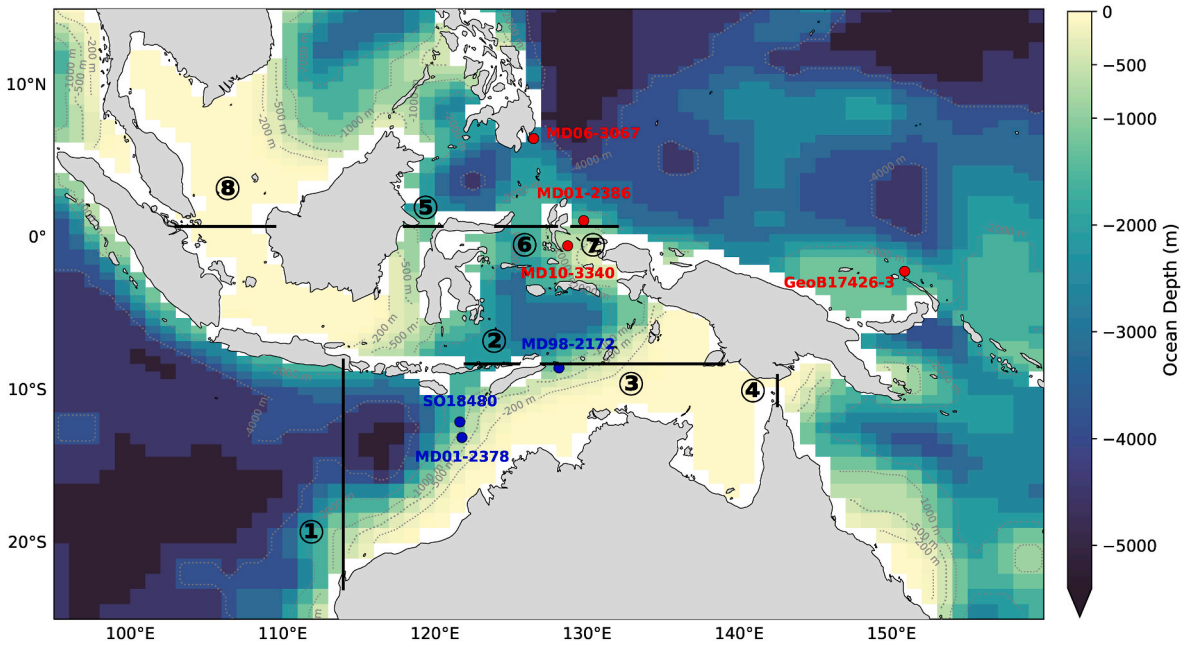


Fig. 2. FGOALS-g2 ocean grid bathymetry and the eight sections defined for volume transport calculations in the Indonesian Seas. The sections are: (1) 114°E transect, (2) Ombai Strait, (3) Timor Passage, (4) Torres Strait, (5) Makassar Strait, (6) Lifamatola Strait, (7) Halmahera Sea, and (8) Karimata Strait. The proxy sites are marked by dots, with red and blue colors indicating sites in the ITF inflow and outflow regions, respectively.

2022; Xu et al., 2006, 2008; Zuraida et al., 2009). For better comparison, we grouped these records into two stacks: one representing the ITF inflow region (Cores MD06-3067, MD01-2386, MD10-3340, GEOB17426-3), and the other representing the ITF outflow region (Cores MD98-2172, SO18480, MD01-2378). We applied a unified calibration to calculate temperatures from Mg/Ca ratios for all sites (Equation (3)) in order to eliminate potential biases arising from the use of different calibrations.

$$\frac{Mg}{Ca}\left(\text{mmol mol}^{-1}\right) = Be^{AT} \tag{3}$$

where A = 0.097, B = 0.260 for *G. ruber*, and A = 0.090, B = 0.328 for *P. obliquiloculata* (Anand et al., 2003; Jian et al., 2022).

3. Results

3.1. Response of volume transport and velocity

Among the eight sections analyzed, substantial increases in volume transport were primarily observed in Sections 1, 2, 6, and 7 (Fig. 3). These sections correspond to the total ITF transport from the Pacific into the Indian Ocean, the transport across the Ombai Strait, the Lifamatola Strait, and the Halmahera Sea, respectively. Relative to the PI control run, the enhancements were: Section 1 by 30.6–35.9% (5.65–6.62 Sv); Section 2 by 52.2–64.6% (6.46–8.00 Sv); Section 6 by 249.0–289.7% (3.61–4.20 Sv); Section 7 by 148.5–276.3% (1.44 Sv–2.68 Sv). Vertically, transport changes were negligible in the upper 354 m but intensified at depths below 354 m, encompassing both the intermediate (354–1021 m) and deep (>1021 m) layers (Fig. 3).

We further analyzed the velocities perpendicular to these sections. In Section 1, the westward velocity intensified mainly at a depth of around 1000 m (Fig. 4A), contributing significantly to the overall transport increase. Sections 2 and 6 exhibited patterns similar to Section 1, with southward velocity enhancements concentrated near 1000 m (Fig. 4B and C). In contrast, in Section 7, the southward velocity increased mainly in the upper ocean (Fig. 4C).

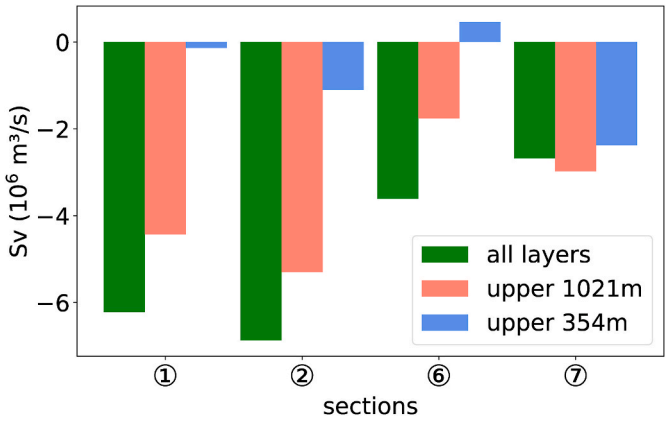


Fig. 3. Changes in mean annual volume transport at 128ka relative to the PI control run. The four sections with the most significant changes in volume transport were: ① the section along 114°E from Australia to Indonesia, ② Ombai Strait, ⑥ Lifamatola Strait, and ⑦ Halmahera Sea. Green, red, and blue bars denote volume transport integrated over all depths, the upper 1021 m, andthe upper 354 m, respectively. Positive values denote northward and eastward transport. The 128 ka experiment is shown as a representative case for brevity.

3.2. Response of temperature

Surface temperatures in the three early LIG experiments (125-130ka) were generally warmer at mid-to high latitudes compared to the PI control run (Fig. 5), with a maximum warming of ~5°C in the southern high latitudes. By contrast, surface temperatures in tropical regions were slightly cooler (decreasing by ~1°C) during the early LIG. These temperature anomalies resulted from positive(negative) annual insolation anomalies at high(tropical) latitudes driven by higher obliquity during the early LIG. This reduced meridional temperature gradient (Fig. 5), known as the polar amplification phenomenon, was also evident from proxy records (Hoffman et al., 2017; Turney et al., 2020; Turney and Jones, 2010). The glacial inception experiment (115ka) showed a

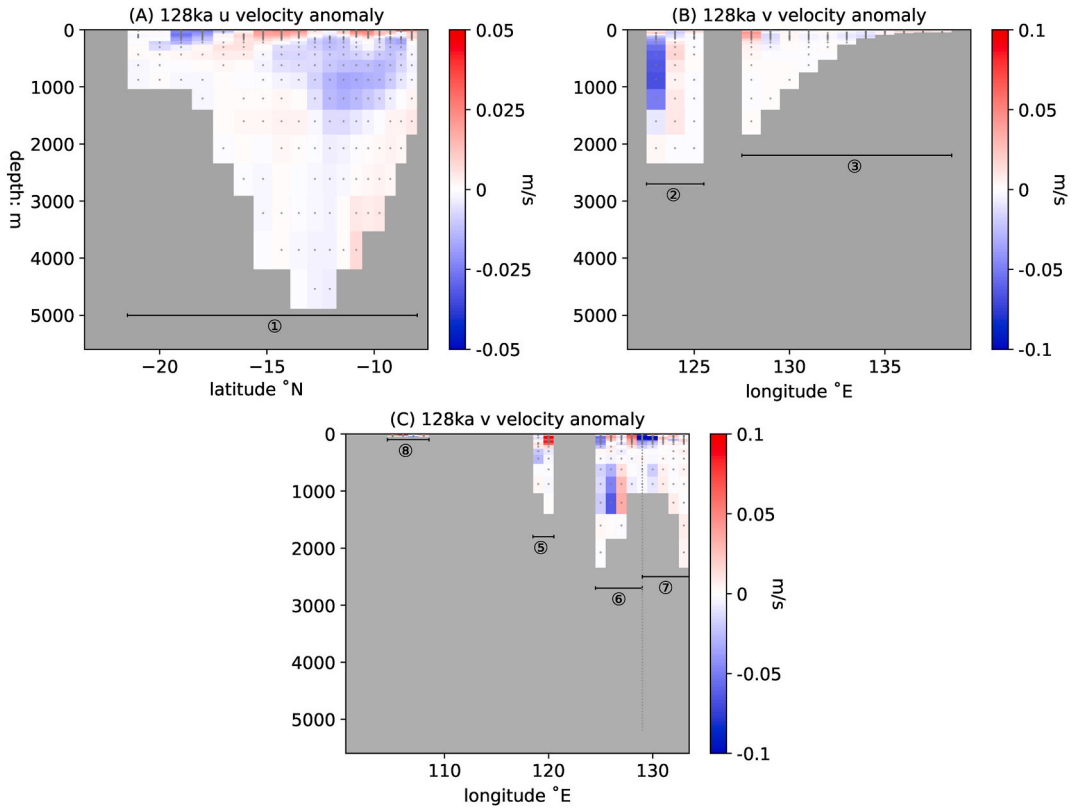


Fig. 4. Changes in mean annual velocity perpendicular to the sections at 128ka relative to the PI control run. (A) Section 1; (B) Sections 2 and 3; (C) Section 5-8. U (zonal) and V (meridional) velocities represent east-west and north-south flows, respectively, with positive values indicating eastward and northward movement. Stippling indicates regions where anomalies are statistically significant at the 95% confidence level (Welch's *t*-test). The 128 ka experiment is shown as a representative case for brevity.

significant Arctic cooling (~5.5°C) coincident with a minimum in Northern Hemisphere summer insolation. Overall, despite prominent regional anomalies, global mean annual surface temperatures in the four experiments were close to pre-industrial values (−0.30°C to 0.44°C). Similar case happens in the CMIP6 *lig127k* simulations, where the multi-model mean change across 17 models was slightly colder (−0.2 ± 0.32°C) compared to the PI control run (Otto-Bliesner et al., 2021). However, proxy records indicate that global temperature exceeded present levels by up to 2°C during the LIG (Turney et al., 2020), highlighting the inability of current models to fully capture the LIG warmth.

Global mean annual SSTs in the LIG experiments were generally warmer than those in the PI control run, with anomalies ranging from 0.20°C to 0.61°C, falling within the uncertainty range of peak LIG global mean annual SSTs compiled by Hoffman et al. (2017). In the tropical Pacific Ocean, a cooling band emerged due to higher obliquity that resulted in reduced mean annual insolation (Fig. 6A). Notably, cooling was more pronounced in the central and central-eastern tropical Pacific compared to the western tropical Pacific, indicating a La Niña-like state. More specifically, we calculated the tropical Pacific SST gradient, defined as the difference in mean SST between the western Pacific (WP; 110°–160°E, 5°S–5°N; L'Heureux et al., 2013) and the Niño 3.4 region (120°–170°W, 5°S–5°N; Trenberth, 1997). During the early LIG (125-130ka), this zonal SST gradient (WP minus Niño 3.4) increased by 0.68 to 1.07°C relative to the PI, indicating a notably enhanced gradient. The 115 ka experiment exhibits a similar but weaker increase of 0.50°C.

This La Niña-like pattern is consistent with previous proxy reconstructions (Pena et al., 2008; S. Zhang et al., 2021). In contrast to surface waters, tropical Pacific subsurface waters warmed, with greater warming in the western Pacific compared to the east (Fig. 6B). This subsurface warming is consistent with reconstructions from planktonic foraminiferal Mg/Ca ratios (Jian et al., 2022; Pena et al., 2008; S. Zhang et al., 2021).

3.3. Response of wind patterns and precipitation

Surface easterlies intensified in the western tropical Pacific Ocean, extending from about 140°E to 160°W (Fig. 6C), indicating an enhanced Pacific Walker Circulation during the early LIG. The strength of the Walker circulation and equatorial easterlies can be inferred from the large-scale SLP contrast across the tropical Pacific (Vecchi et al., 2006; Kosovelj and Zaplotnik, 2023). The zonal SLP gradient (Niño 3.4 minus WP) increased by 0.84 to 1.08 hPa during the early LIG. Similarly, the 115 ka experiment shows an increased SLP gradient, though with a smaller magnitude (0.45 hPa). Because this zonal pressure difference directly drives the near-surface winds, its significant increase demonstrates both a strengthened Walker circulation and intensified equatorial easterlies. Consistently, PMIP4 *lig127k* simulations also suggested a broadened and strengthened Pacific Walker Circulation (Ma et al., 2024). These enhanced easterlies led to increased precipitation over the Maritime Continent (Fig. 6D). Pollen records from West Java (van der Kaars and Dam, 1997) and sediment records from the Halmahera Sea (Dang et al., 2015) reveal a more humid Maritime Continent during the early LIG, corroborating the model results.

3.4. Evidence from proxy records

The ΔT records of the ITF inflow and outflow regions over the last 150 kyr are shown in Fig. 7, which serve as proxies for the DOT. During the early LIG, both regions experienced a pronounced decrease in ΔT, indicating a deepening of the thermocline. Subsequently, towards the end of the LIG, ΔT shifted to the opposite extreme in both regions,

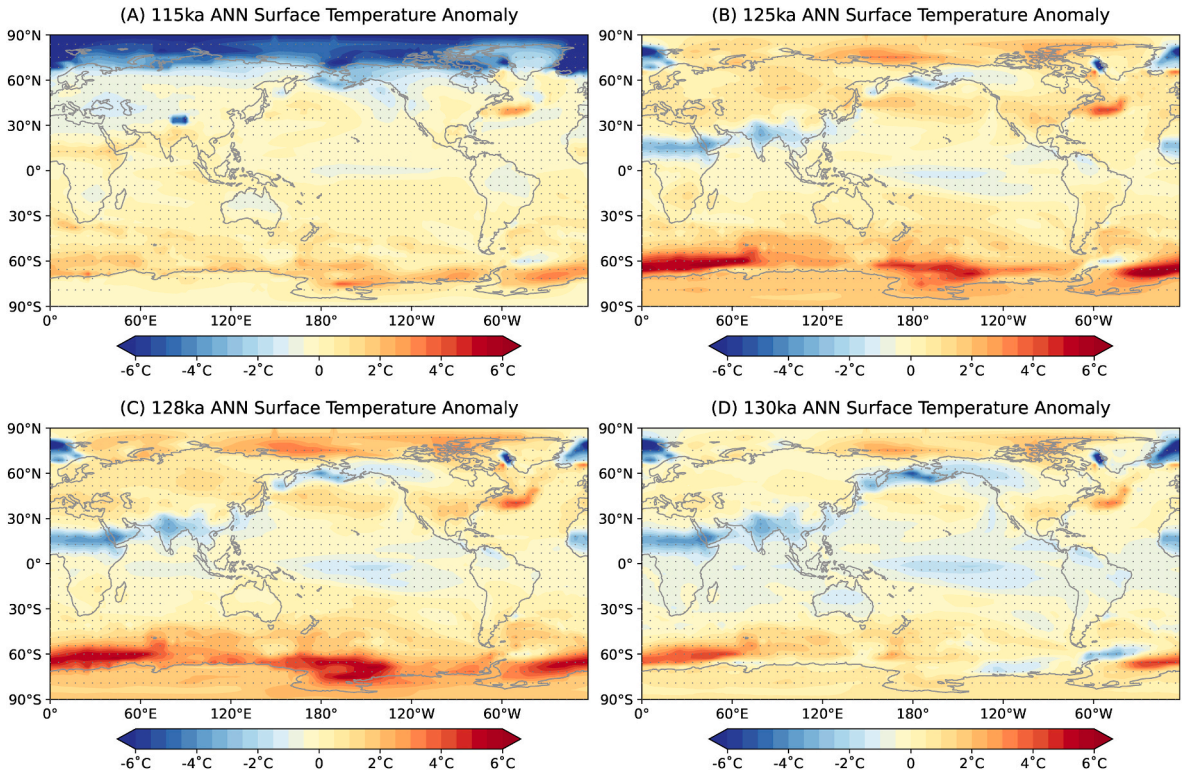


Fig. 5. Changes in mean annual surface temperature at (A) 115ka, (B) 125ka, (C) 128ka, and (D) 130ka relative to the PI control run. As results are similar across the three early LIG snapshots (125-130ka), the 128 ka experiment is used as a representative case for analysis to maintain simplicity. Stippling indicates regions where anomalies are statistically significant at the 95% confidence level (Welch's *t*-test), subsampled at every other grid point for visual clarity.

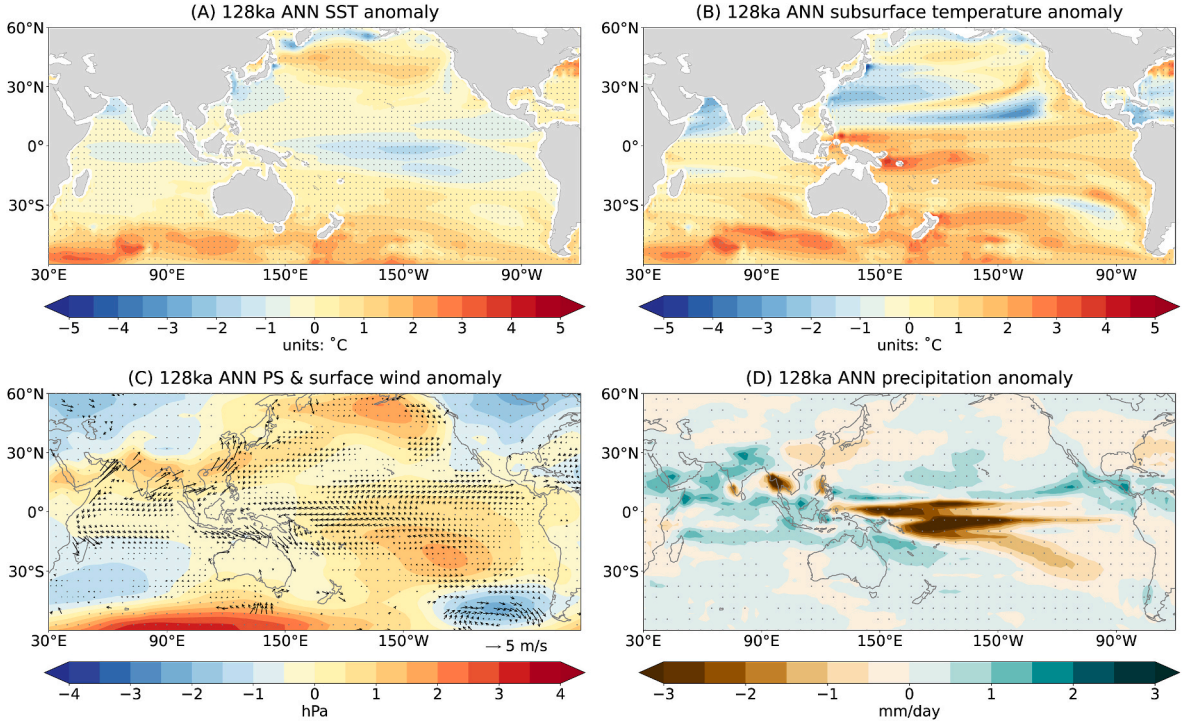
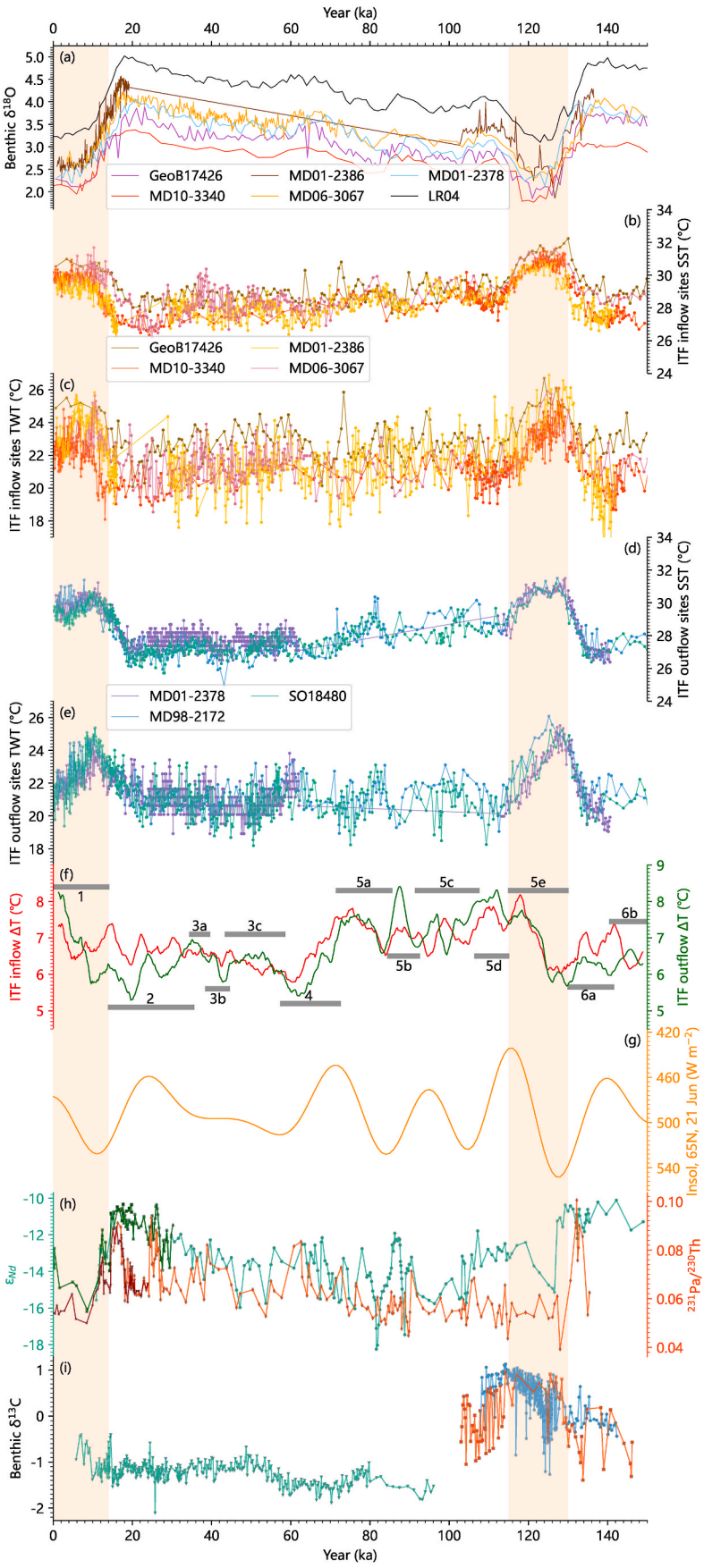


Fig. 6. Changes in mean annual (A) sea surface temperature, (B) subsurface temperature (178.4 m depth), (C) surface pressure and wind pattern, and (D) precipitation at 128ka relative to the PI control run. Stippling indicates regions where anomalies are statistically significant at the 95% confidence level (Welch's *t*-test), subsampled at every other grid point for atmospheric fields and every third grid point for ocean fields for visual clarity. Only wind vectors passing the same significance test are shown.

suggesting a shoaling of the thermocline. Similarly, the DOT in the outflow region was deep during the early Holocene and shoaled over



(caption on next page)

Fig. 7. Proxy records. (A) Benthic $\delta^{18}O$ records (Bolliet et al., 2011; Jian et al., 2020; Holbourn et al., 2005; Hollstein et al., 2020) from ITF region core sites compared to the LR04 global stack (Lisiecki and Raymo, 2005). The chronologies of different sites are consistent, allowing for direct comparison without age tuning. (B-E) SSTs and thermocline water temperatures (TWTs) derived from Mg/Ca ratios of *G. ruber* and *P. obliquiloculata* from ITF inflow region cores (B–C: GeoB17426, MD10-3340, MD01-2386, and MD06-3067; Hollstein et al., 2020; Jian et al., 2020, 2022; Bolliet et al., 2011) and ITF outflow region cores (D-E: MD01-2378, MD98-2172, and SO18480; Xu et al., 2008, 2006; Zuraida et al., 2009; Ding et al., 2023; Jian et al., 2022). (F) $\Delta T_{SST-TWT}$ for ITF inflow (red) and outflow (green) regions. The curves represent the averages of the respective core sites listed in (B–C) and (D–E). (G) Insolation at 65°N on 21 June (Laskar et al., 2004). (H) ε_{Nd} records (green lines) from North Atlantic core sites ODP 1063 (circle, Böhm et al., 2015; triangle, Gutjahr and Lippold, 2011) and GGC6 (square, Roberts et al., 2010). $^{231}Pa/^{230}Th$ records (orange lines) from North Atlantic core sites ODP 1063 (diamond, Böhm et al., 2015; inverted triangle, Lippold et al., 2009) and GGC5 (star, McManus et al., 2004). (I) Benthic $\delta^{13}C$ records from North Atlantic core sites ODP 1063 (orange, Deaney et al., 2017) and MD03-2664 (blue, Galaasen et al., 2014), and North Pacific core U1340 (green, Schlung et al., 2013). Orange shading marks the Holocene and LIG intervals.

time. The DOT in the ITF inflow region during the Holocene appears to have undergone a similar transition from deep to shallow, albeit with a smaller magnitude. The relative change in DOT between the inflow and outflow regions was not clear probably due to proxy limitations. Overall, it is evident that both regions experienced a prominent deepening of the thermocline during the early LIG relative to the PI.

4. Discussion

4.1. La Niña–like conditions and strengthening of the ITF during the early LIG

During the early LIG, the model results indicate a La Niña-like mean state, characterized by an intensified zonal SST gradient and enhanced tropical Pacific easterlies. Although the tropical Pacific generally cooled due to higher obliquity, the central and central-eastern regions experienced more significant cooling than the western region (Fig. 6A). This enhanced SST gradient strengthened the Walker circulation and intensified the surface easterlies in the western tropical Pacific (Fig. 6C). As a result, relatively warmer water accumulated in the Western Equatorial Pacific (WEP), leading to a deepened thermocline there (Fig. 6B). The increased SST gradient and intensified easterlies acted as a positive feedback, reinforcing one another and resulting in a La Niña-like state.

The thermocline depth in the tropical Pacific is closely related to the ENSO state. Using the 20°C isotherm to represent thermocline depth, we observe a deepening of the thermocline in both the WEP and the Eastern Equatorial Indian Ocean (EEI), with significantly greater deepening in the WEP than in the EEI during the early LIG (Fig. 8). Since SSH is closely tied to thermocline depth in the tropics (Rebert et al., 1985)—because a deeper thermocline implies a thicker, warmer, and lower-density layer above it—this asymmetric thermocline deepening resulted in a greater SSH increase in the WEP compared to the EEI. Thus, the SSH gradient between the WEP (125°–135°E, 4°–6°N) and the EEI (105°–132°E, 13°–8°S; regions defined based on Shinoda et al., 2012; Wyrtki, 1987) increased by 0.018 to 0.046 m during the early LIG. Today, the pressure gradient maintained by relatively higher SSH in the WEP and lower SSH in the EEI drives the ITF (Wyrtki, 1987). During the early LIG, this pressure gradient was intensified by the asymmetric increase in SSH. Consequently, the steeper pressure gradient drove greater water transport through the Indonesian archipelago.

The La Niña-like state and the enhancement of the ITF during the LIG are well-supported by proxy data. The TWT and $\Delta T_{SST-TWT}$ in the WEP and the ITF region are strongly associated with the ENSO state on different timescales. When El Niño-like activity increases, sea surface and thermocline temperatures decrease in both the WEP and the ITF region, resulting in an increase in $\Delta T_{SST-TWT}$ (Jian et al., 2020; P. Zhang et al., 2018, 2021; S. Zhang et al., 2021). During the early LIG, proxy records show that TWT increased and $\Delta T_{SST-TWT}$ decreased in the WEP (Bolliet et al., 2011; Hollstein et al., 2020; Jian et al., 2020, 2022; S. Zhang et al., 2021). This is consistent with our model results, pointing to a deepening of the thermocline there and a La Niña-like state. Previous studies have found that the southern–northern Makassar Strait thermocline temperature gradient decreases during a La Niña-like mean state, indicating enhanced ITF transport, and vice versa during an El Niño-like mean state (Fan et al., 2018). These same effects of ENSO on the Makassar Strait throughflow also operate on centennial to millennial timescales (Fan et al., 2013). Using the TWT gradient between the Java-Sumatra upwelling region and the Lombok region as an indicator of ITF strength, Pang et al. (2021) also found an ITF enhancement during the LIG.

Our collected proxy records show that $\Delta T_{SST-TWT}$ decreased in both the ITF inflow and outflow regions during the early LIG (Fig. 7), indicating a deepening of the thermocline in both regions, consistent with the model results (Fig. 8). However, the magnitudes of $\Delta T_{SST-TWT}$ were

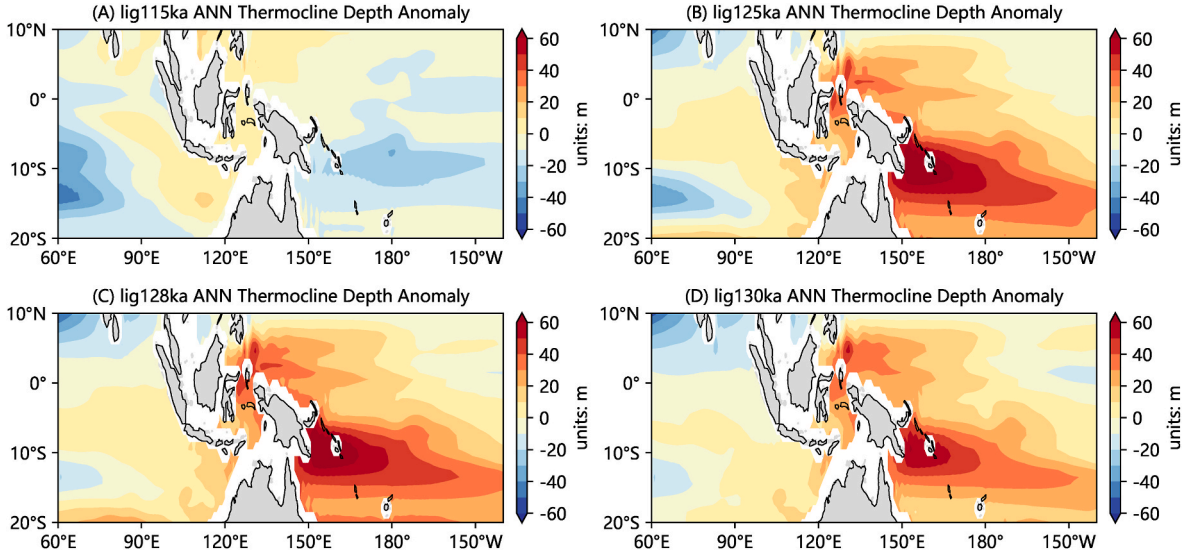


Fig. 8. Changes in mean annual thermocline depth at (A) 115ka, (B) 125ka, (C) 128ka, and (D) 130ka relative to the PI control run.

quite similar between the ITF inflow and outflow regions during the LIG, which was inconsistent with the asymmetrical increase in thermocline depth observed in the model results. This mismatch likely reflects limitations in current proxy records rather than a true absence of asymmetry.

Other independent lines of evidence further support our findings. Previous modeling studies have also shown an enhanced Pacific Walker Circulation (Ma et al., 2024) and a decreased ENSO index (Clement et al., 1999; Clement and Cane, 1999) during the early LIG. Pollen records from West Java (van der Kaars and Dam, 1997) and terrigenous sediment input from the Halmahera Sea (Dang et al., 2015) indicate increased precipitation in the Maritime Continent, suggesting an enhancement of tropical easterly winds, consistent with our model results. Thus, we propose that the increased SST gradient and intensified easterlies under a La Niña-like state acted as a positive feedback loop. This raised SSH in the WEP, steepened the pressure gradient between the WEP and EEI, and hence drove the enhanced ITF volume transport during the early LIG.

4.2. Roles of wind and Pacific deep upwelling in enhancing ITF transport during the LIG

We computed the transport using Godfrey's Island Rule to further assess the role of winds in modulating the ITF during the LIG (Godfrey, 1989). Godfrey's Island Rule elegantly links wind stress to ITF transport by leveraging Sverdrup dynamics in a constrained geometry, assuming that the ocean is stagnant below some depth Z (Equation (4)):

$$T_w = \frac{1}{f_N - f_S}\oint_{ABCD}\frac{\tau^l}{\rho_0}dl \tag{4}$$

where T_w represents the ITF transport calculated by the Island Rule; f is the Coriolis parameter; f_N and f_S are the Coriolis parameters at the northern and southern ends of Australia; ABCD is the integration path; τ^l is the along-route wind stress; and ρ_0 is the mean water density. This formula computes the net transport around the island using only wind stress along the closed path that encloses the island.

Feng et al. (2017) proposed that while vertical velocities at Z are very small, their basin-integrated effect can significantly influence the ITF. They therefore amended the Island Rule to include deep upwelling across the Pacific (Equation (5)):

$$T_{itf} = \frac{1}{f_N - f_S}\oint_{ABCD}\frac{\tau^l}{\rho_0}dl + \iint_{Pacific} w_Z ds \tag{5}$$

where T_{itf} represents the total ITF volume transport; and w_Z denotes the vertical velocity at the chosen reference depth Z. The first term corresponds to the traditional Island Rule, whereas the second term represents the upwelling integrated over the Pacific basin north of 38°S at the reference depth Z. Here Z = 2000m was adopted as the reference depth, which lies below the sill depth of the Indonesian seas and ensures volume conservation.

Based on Godfrey's Island Rule, estimates from the LIG experiments indicate that wind-driven ITF transport increased significantly during the La Niña–like early LIG and decreased slightly at 115 ka, consistent with the analysis above. However, the magnitude of the wind-driven increase was only 2.12–2.64 Sv in the 125-130 ka experiments, substantially smaller than the total ITF increase of 5.65–6.62 Sv. Moreover, the 115 ka experiment—despite exhibiting an El Niño–like mean state—showed an ITF increase comparable in magnitude to that of the 125–130 ka experiments. Together, these results imply that winds alone could not account for the full ITF enhancement during the LIG, indicating that additional mechanisms were at play.

We further applied the amended Island Rule (Feng et al., 2017) and found that Pacific deep upwelling at 2000 m increased markedly in the early LIG experiments (Table 2), by 1.32–2.69 Sv—roughly the same magnitude as the wind contribution (2.12–2.64 Sv). At 115 ka, the ITF increase was primarily driven by enhanced Pacific upwelling (4.25 Sv). From 130 ka to 115 ka, Pacific upwelling exhibited an increasing trend, which helped explain why the 115 ka experiment maintained a total ITF transport similar to early LIG values, despite the absence of a pronounced La Niña–like state.

Table 2
Quantified wind stress and Pacific upwelling contributions to ITF transport (unit: Sv).

	PI	115ka	125ka	128ka	130ka
Island Rule	9.81	9.59	11.93	12.45	11.97
Pacific upwelling	12.62	16.87	15.31	14.45	13.94
Total ITF volume transport	18.46	25.04	25.08	24.68	24.11

Based on mass balance, the enhanced Pacific deep upwelling during the LIG was likely driven by enhanced NADW formation and an intensified AMOC. We calculated vertical volume transports across 2000 m in different basins (Table 3). In the Atlantic north of 30°N, enhanced downwelling of 8.33–9.67 Sv contributed substantially to NADW formation. Another region of increased deep water formation was the Pacific Antarctic sector south of 38°S (1.26–4.74 Sv). The excess downward transport below 2000 m was primarily balanced by upwelling in the Indian Ocean (2.99–4.04 Sv), the Pacific Ocean (1.32–4.25 Sv), and the Atlantic Antarctic sector (2.38–2.97 Sv) during the LIG (Table 3). Notably, the stronger Pacific upwelling at 115 ka relative to 125–130 ka was largely driven by enhanced downwelling in the Pacific Antarctic sector (2.53–3.48 Sv). This explains why the 115 ka experiment exhibited ITF transport comparable to the 125–130 ka experiments despite the absence of a La Niña-like state.

Model results also show that the AMOC generally strengthened during the LIG (Fig. 9). The AMOC index—defined as the maximum Atlantic meridional stream function below 500 m between 80°N and 34°S (Hu et al., 2021)—increased by about 0.47–1.37 Sv (Table 4). In contrast, the AABW index—defined as the maximum strength of the global meridional overturning stream function south of 60°S (Lago and England, 2019)—generally decreased during the LIG by about 0.39–0.89 Sv. A reduction in AABW formation was also evident in Fig. 10, where the lower Global Meridional Overturning Circulation (GMOC) cell shoaled and weakened in magnitude. The AABW did not penetrate as far north during the LIG as in the PI. Overall, the GMOC index—defined as the largest absolute value of the global meridional stream function below 500 m between 70°S and 20°N (Yuan et al., 2024)—increased by 0.92–2.08 Sv during the LIG, with the maximum values located in the upper clockwise cell. Notably, although the AABW weakened globally during the LIG, it exhibited regional differences: it weakened in the Atlantic but shoaled and strengthened in the Pacific (Fig. 11).

The strengthened AMOC and increased formation of NADW during the LIG are supported by proxy records from the North Atlantic. It is well known that NADW has significantly lower ε_{Nd} and $^{231}Pa/^{230}Th$ than the southern-sourced waters. During the LIG and the Holocene, both ε_{Nd} and $^{231}Pa/^{230}Th$ jumped from their preceding glacial maximum values to prominently lower values, indicating the dominance of NADW over southern-sourced waters and a strengthened AMOC (Fig. 7H; Böhm et al., 2015; Roberts et al., 2010). Additionally, because

Table 3
Vertical volume transports across 2000m at different regions (unit: Sv).

	PI	115ka	125ka	128ka	130ka
Pacific Upwelling	12.62	16.87	15.31	14.45	13.94
North Atlantic (>30°N) Upwelling	−13.05	−21.38	−22.22	−22.72	−22.31
Indian Ocean Upwelling	2.83	5.86	5.82	6.87	6.28
Pacific Antarctic sector Upwelling	−7.28	−12.02	−9.49	−8.62	−8.54
Atlantic Antarctic sector Upwelling	6.04	8.42	8.92	8.90	9.01

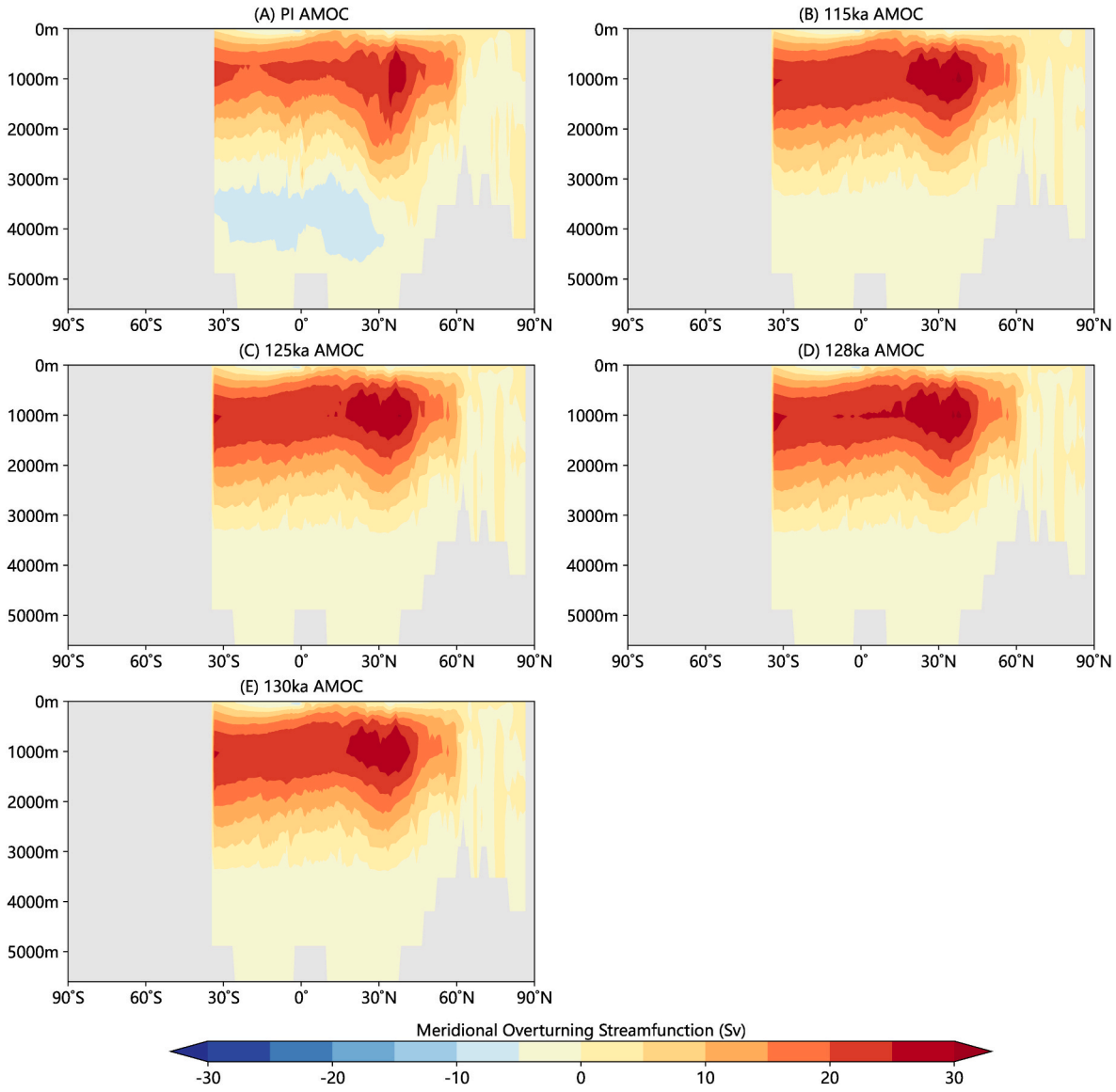


Fig. 9. AMOC during PI control and LIG simulations (units: Sv).

Table 4
Indices of key overturning cells (unit: Sv).

	PI	115ka	125ka	128ka	130ka
GMOC	26.00	26.92	28.08	27.93	27.05
AMOC	29.36	30.73	30.17	30.55	29.83
AABW	5.70	5.31	4.81	5.18	4.98

northern-sourced waters are better ventilated and exhibit higher $\delta^{13}C$ than the southern-sourced waters, the increase in benthic $\delta^{13}C$ in the North Atlantic during the LIG further points to enhanced NADW formation (Fig. 7I; Deaney et al., 2017; Galaasen et al., 2014).

Notably, following the CMIP5 protocol, the topography and coastlines in our LIG simulations were fixed to PI conditions. Consequently, the 1.2–5.3 m GMSL rise characteristic of the early LIG period (Dyer et al., 2021) was not explicitly resolved in our model setup. Recent modeling evidence by Zhang et al. (2023) demonstrates that even a modest rise in GMSL can substantially reorganize global atmospheric and oceanic circulations. Specifically, their sensitivity experiments revealed that higher sea levels deepen ocean gateways, thereby facilitating enhanced inter-basin water mass exchanges. Consistent with this, Sarim et al. (2025) noted that interglacial sea-level rise broadened these gateways, resulting in an intensified ITF. In this context, a higher sea level during the LIG would deepen the Indonesian seas. Therefore, incorporating realistic LIG sea-level rises into our experiments would potentially amplify the simulated ITF intensification. Concurrently, this GMSL rise also impacts the AMOC. Higher GMSL drives more Pacific water flow into the Arctic via a deepened Bering Strait, which acts to weaken the AMOC. For instance, in the 5 m LIG GMSL rise experiment carried out by Zhang et al. (2023), AMOC strength decreased by 2%. However, for GMSL increases of less than 5 m, water transport into the North Atlantic via the Bering Strait remains relatively small, suggesting a correspondingly minor impact on AMOC strength (Zhang et al., 2023).

4.3. *Comparison of ITF behaviour between LIG and future global warming*

Under the future global warming scenario of approximately 4.4°C (SSP5-8.5; IPCC, 2021), the Walker Circulation is consistently projected to slow down (Peng et al., 2023; Vecchi et al., 2006), accompanied by equatorial Pacific warming centered in the east (Peng et al., 2023). These wind and temperature patterns contrast with those of the early LIG. While wind pattern changes in the 21st century favor decreased ITF

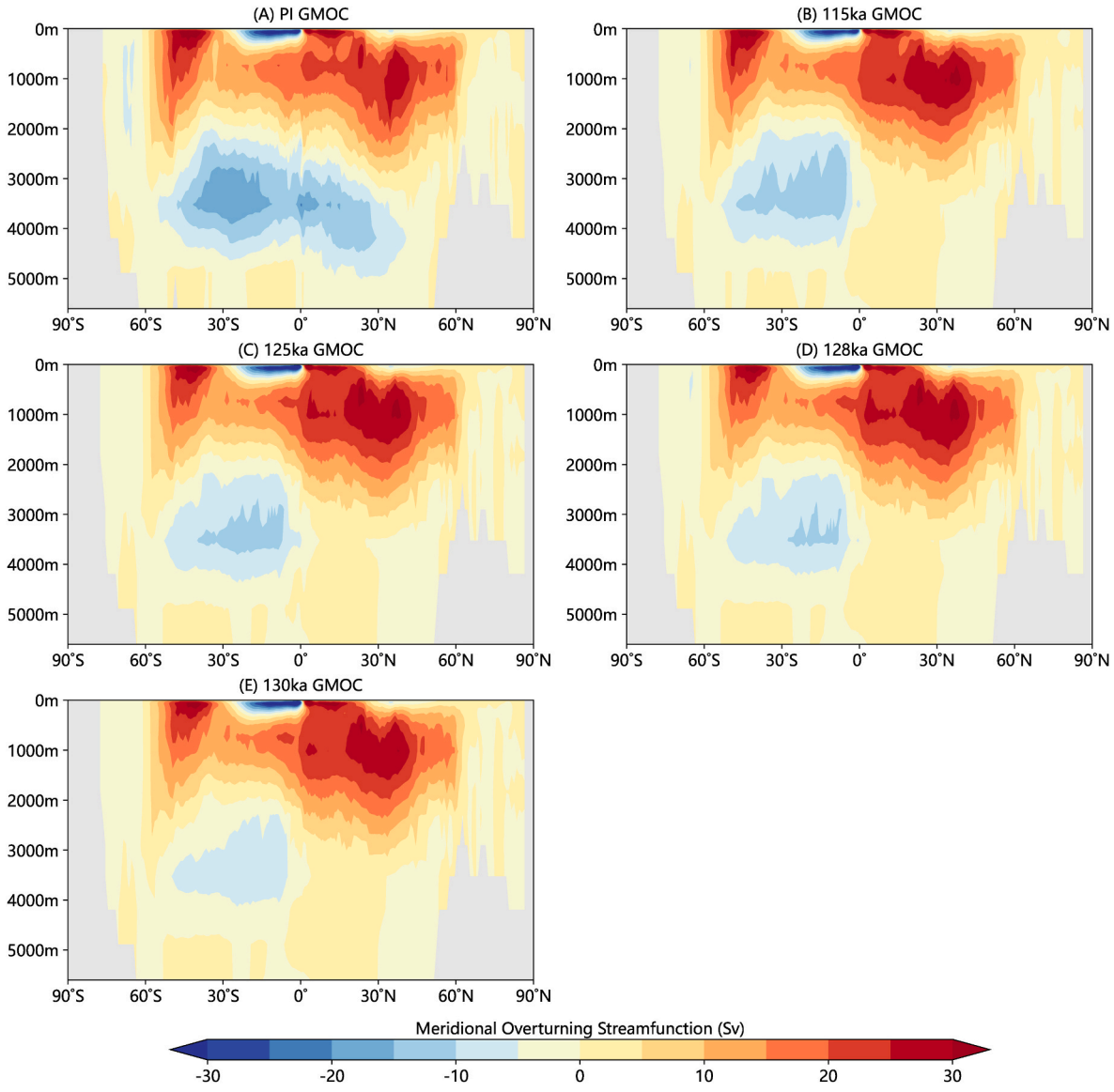


Fig. 10. GMOC during PI control and LIG simulations (units: Sv).

transport, they favored enhanced ITF transport during the early LIG. Interestingly, for both periods, wind pattern changes alone could not explain all the ITF changes (Sen Gupta et al., 2016; Feng et al., 2017; Stellema et al., 2019; Sun and Thompson, 2020; Peng et al., 2023). Consequently, several mechanisms have been proposed, such as reduced Pacific deep upwelling and the remote influence of the AMOC.

Sen Gupta et al. (2016) found a significant correlation between the mean twentieth-century upwelling and mean ITF transport across 22 CMIP5 climate models. Feng et al. (2017) further incorporated Pacific upwelling into an amended Island Rule and found that wind pattern changes contributed to only roughly one-third of the ITF decrease, with the remaining two-thirds driven by reduced Pacific deep upwelling. Conversely, while Pacific deep upwelling is projected to decline in the 21st century, it increased during the LIG, leading to opposing ITF changes between the two periods.

Sun and Thompson (2020) and Peng et al. (2023) showed that the weakened AMOC due to greenhouse gas forcing would induce interface deepening (Sun and Thompson, 2020) and elevate dynamic height (Peng et al., 2023) in the North Atlantic. These signals propagate to the Indian Ocean via wave processes. Both signals ultimately lead to elevated Indian Ocean SSH, which reduces the Pacific-Indian SSH gradient and weakens the ITF. Conversely, the AMOC was strengthened during the LIG which led to opposite ITF changes. From the perspective of mass conservation, we further attribute the increased Pacific deep upwelling to increased NADW formation, which resulted in increased ITF transport, in contrast to future projections.

The distinct driving mechanisms behind climate warming in the future and the LIG play a decisive role in generating these opposing trends in wind patterns and Pacific upwelling. While future global warming is primarily driven by increased greenhouse gas concentrations, the climate changes during the LIG were principally caused by changes in orbital parameters.

5. Conclusions

By analysing a set of LIG simulations performed with FGOALS-g2 following the PMIP protocol, we found a 30.6%–35.9% increase in annual mean ITF transport during the early LIG relative to the PI. This enhancement is closely correlated with a La Niña-like mean state and strengthened Pacific deep upwelling.

During the early LIG, the increased SST gradient between the WEP and Eastern Equatorial Pacific and intensified easterlies in the tropical

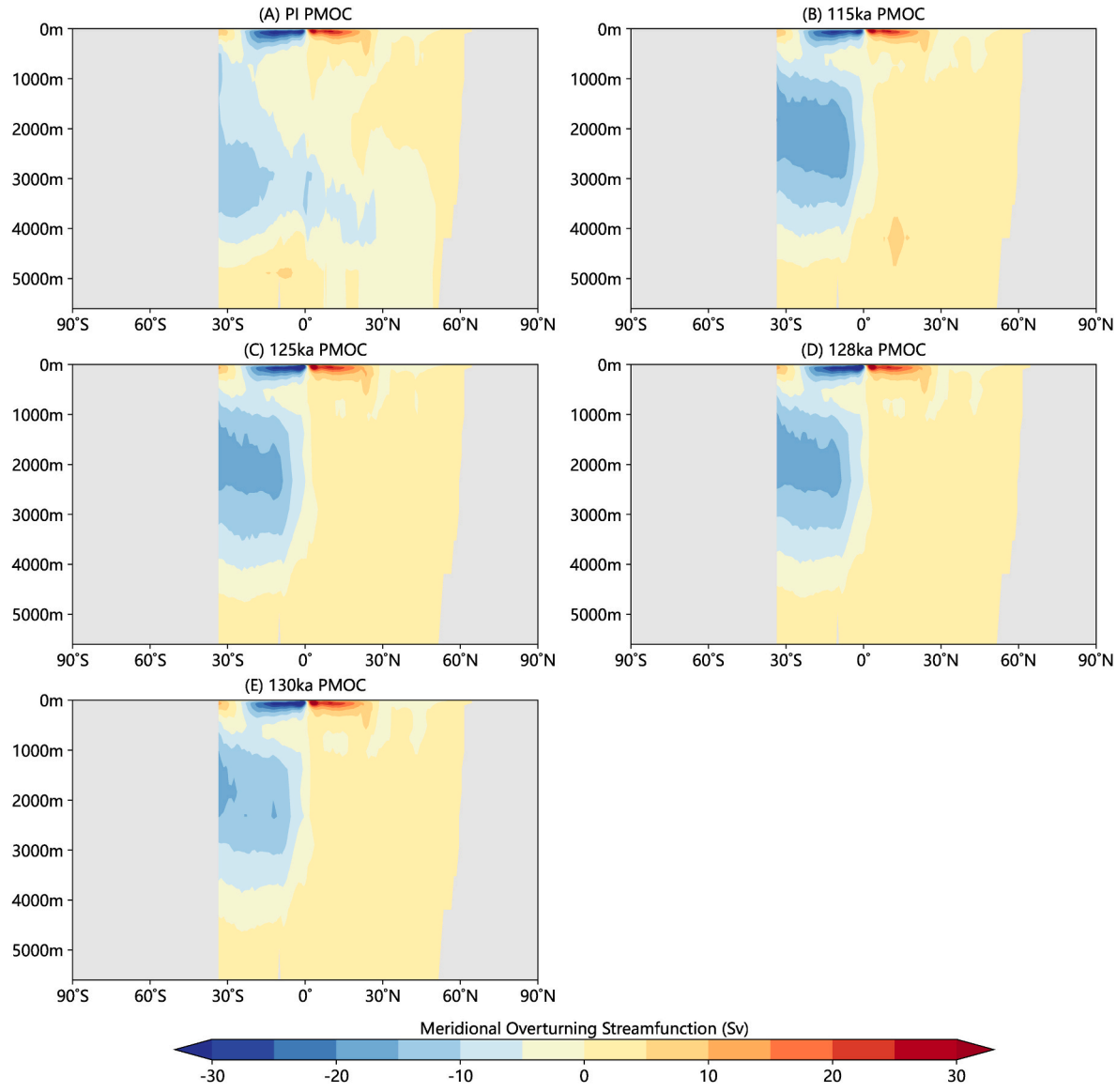


Fig. 11. Pacific Meridional Overturning Circulation (PMOC) during PI control and LIG simulations (units: Sv).

Pacific acted as a positive feedback loop, resulting in a La Niña-like state. As a result, sea levels rose in the western Pacific, leading to an asymmetric increase in SSH between the WEP and the EEI. This increased the pressure contrast between the two basins and subsequently strengthened ITF volume transport. Proxy records show that the TWT increased and $\Delta T_{SST-TWT}$ decreased in the WEP, pointing to a deepening of the thermocline and a La Niña-like state, which is consistent with the model results. Pollen records and marine sediments indicate increased precipitation in the Maritime Continent, suggesting an enhancement of tropical easterly winds, further supporting the model results.

The wind-driven component contributed approximately 2.12–2.64 Sv to ITF transport during the early LIG. Concurrently, enhanced Pacific deep upwelling, arising from a strengthened AMOC and increased NADW formation, contributed roughly 1.32–2.69 Sv. Proxy records from North Atlantic deep-sea cores corroborate the enhanced AMOC and increased NADW formation during the LIG, aligning with model results. At 115 ka, the ITF increase was dominantly driven by increased Pacific upwelling (4.25 Sv), highlighting the important role of deep-ocean processes in modulating ITF variability.

Although the climate was warmer in both the LIG and the 21st century, wind forcing and AMOC changes exhibited opposite trends due to distinct forcing mechanisms. Future global warming is driven by increased greenhouse gases and is characterized by an El Niño-like mean state and a weakened AMOC. In contrast, early LIG warmth was driven by orbital parameters, featuring a La Niña-like mean state and a strengthened AMOC. These contrasts yield opposite ITF responses between the LIG and future scenarios, indicating that in terms of the ITF response, the LIG might not be an appropriate analogue for future climate warming.

Author contributions

S. Wei: Conceptualization, Methodology, Writing-Original Draft. W. Zheng: Resources, Data Curation, Conceptualization. J. Du: Conceptualization, Validation. J. Wei: Conceptualization, Validation. Y. Yu: Conceptualization, Methodology, Formal analysis, Validation. J. Tian: Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. (All authors approved the final manuscript).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.quascirev.2026.110156.

Data availability

The FGOALS-g2 LIG simulation outputs presented in this paper are available on ScienceDB at https://doi.org/10.57760/sciencedb.42716. The PI control simulation data are available via ESGF (CMIP5).

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